

Service Design

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Service Design

1. Principles and Theories of Services
2. Principles and Theories of Design
3. Principles and Theories of Service Design
4. Tools and Methods of Service Design



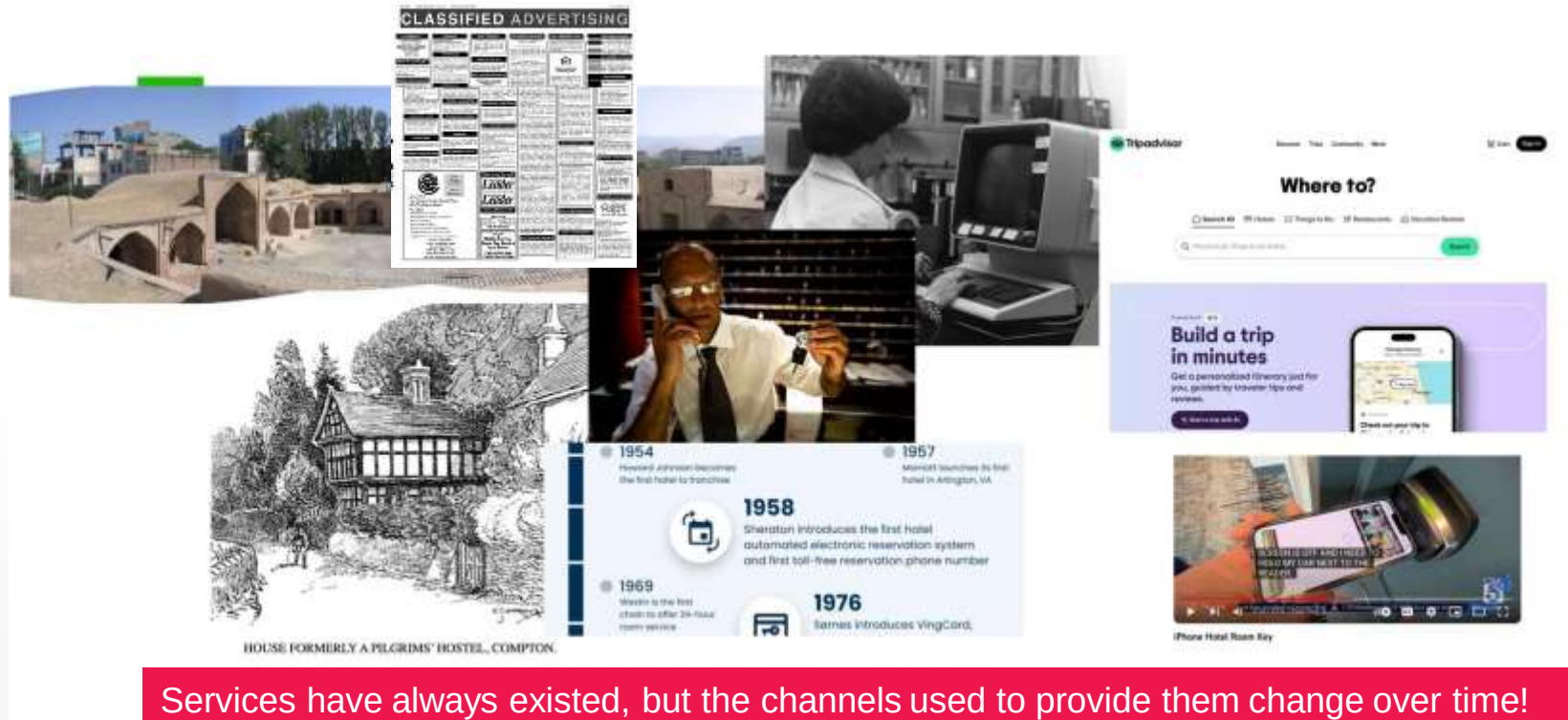


Chapter 1

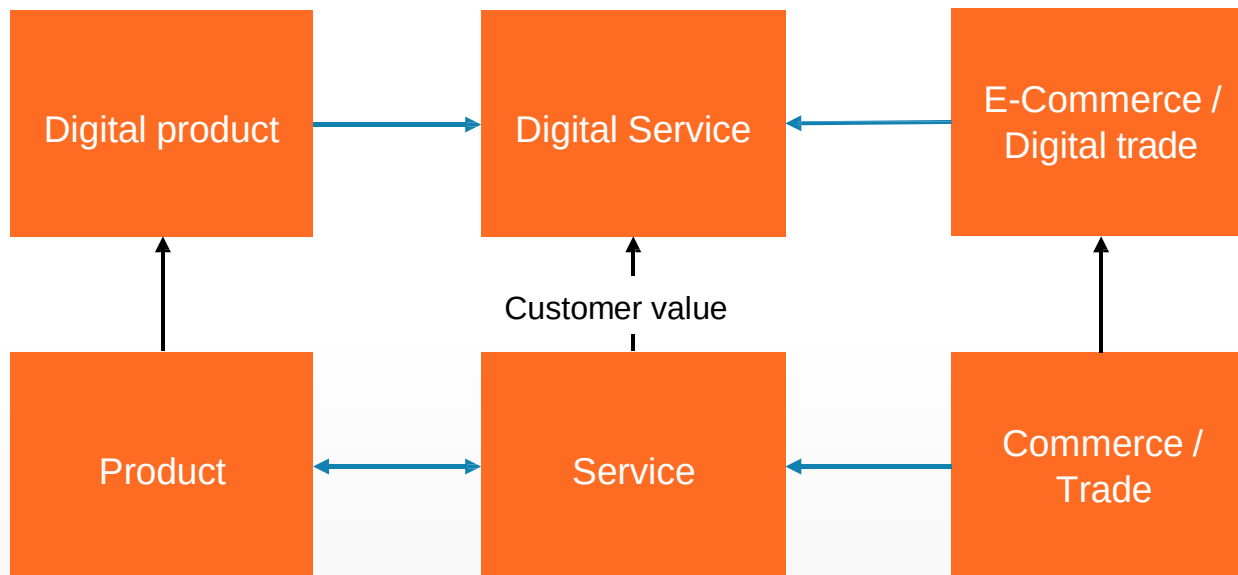
Principles and theories of services

Section: Digital Services and Software

Logging services through time

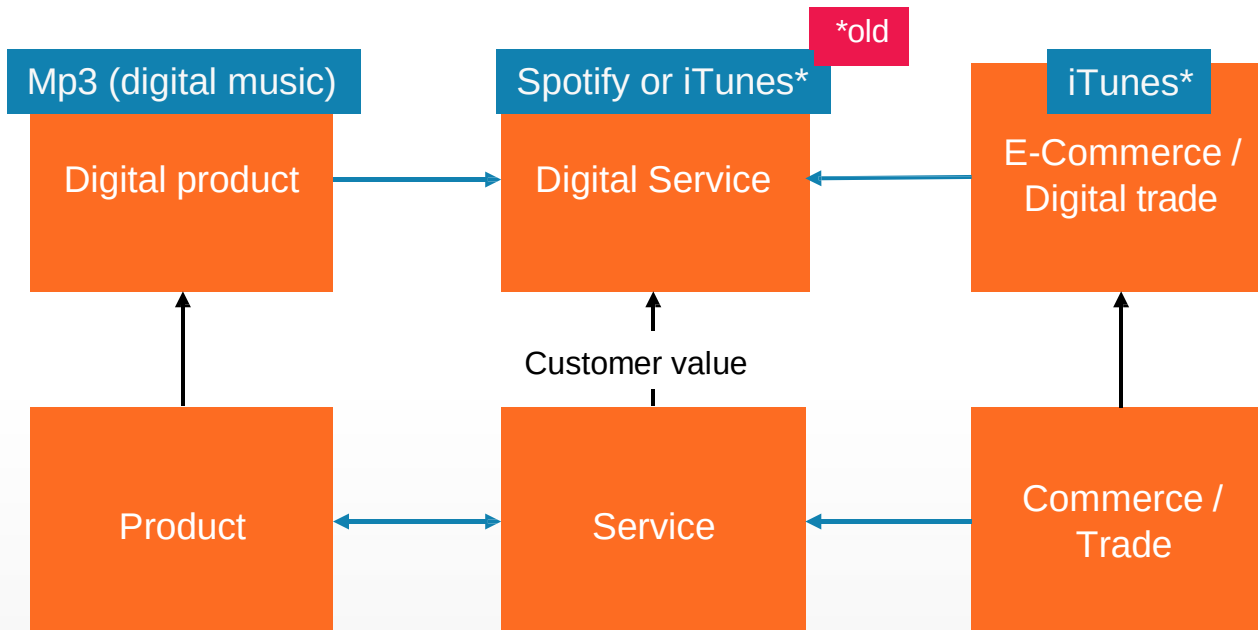


Digital service



Digital service: When the service value chain is digital

Digital service - Example



Another example: eBooks, kindle, Amazon

Servitization of software



Software as a product



Software as a Service (SaaS)



Enabler: Software as service

- **Cloud computing:** Cloud computing enables the scalability of software at low cost, making it feasible to provide software services to a large user base at low costs.
- Software as service enables many different cost models (subscription, cost per usage, etc.)
- **Bandwidth:** Availability of bandwidth enabling connecting almost anywhere at any time.
- **Modular software:** Modular design of service enabling the decoupling of functionality and configuration and composition of services as preferred by users

Container and microservices



Software ecosystems

Software as a Service (SaaS)

DEF

Software as a service (or SaaS) is a way of delivering applications over the Internet—as a service. Instead of installing and maintaining software, you simply access it via the Internet, freeing yourself from complex software and hardware management (*Salesforce*)

DEF

Software as a service (SaaS) allows users to connect to and use cloud-based apps over the Internet. Common examples are email, calendaring, and office tools (*Microsoft*)

DEF

Software as a Service (SaaS) is a software licensing and delivery model in which software is licensed on a subscription basis and is centrally hosted on cloud. Consumers do not need to manage the software architecture, software maintenance and the underlying infrastructure. End users access the SaaS application using a thin client like web browser (*Alibaba*)



<https://www.alibabacloud.com/de/knowledge/what-is-paas>

Software as a Service (SaaS)

Characteristics	Traditional software	SaaS
Ownership	Buying of software	Renting of software without taking ownership
Pricing model	Upfront investments and costs for local installation/maintenance incl. licenses	Pay-per-use or pay-per-period
IT function	Buys, installs, develops, implements and maintains their own software	Subscribe, plug in and use: no need for an IT function and no concern of updates
Expertise needed	In-house software expertise needed for control and maintenance	Usage expertise required

IaaS and PaaS

DEF

Platform as a service (PaaS) is a form of [cloud computing](#) that provides a platform allowing consumers to develop and deploy applications and services over Wide Area Network (WAN) or the Internet. The consumer does not manage or maintain the underlying platform software, middleware and cloud infrastructure.

DEF

Infrastructure as a service (IaaS) is a form of [cloud computing](#) based on automation and orchestration technologies to on-demand provision and manage virtualized infrastructure resources such as server, storage and network over a wide area network (WAN) to the consumer. The consumer does not manage or control the underlying cloud infrastructure but has control over their resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications.



<https://www.alibabacloud.com/de/knowledge/what-is-paas>

KM: The software industry starts to like the term service

Applications data – Apps, Data

Runtime - Docker

Middleware - Software

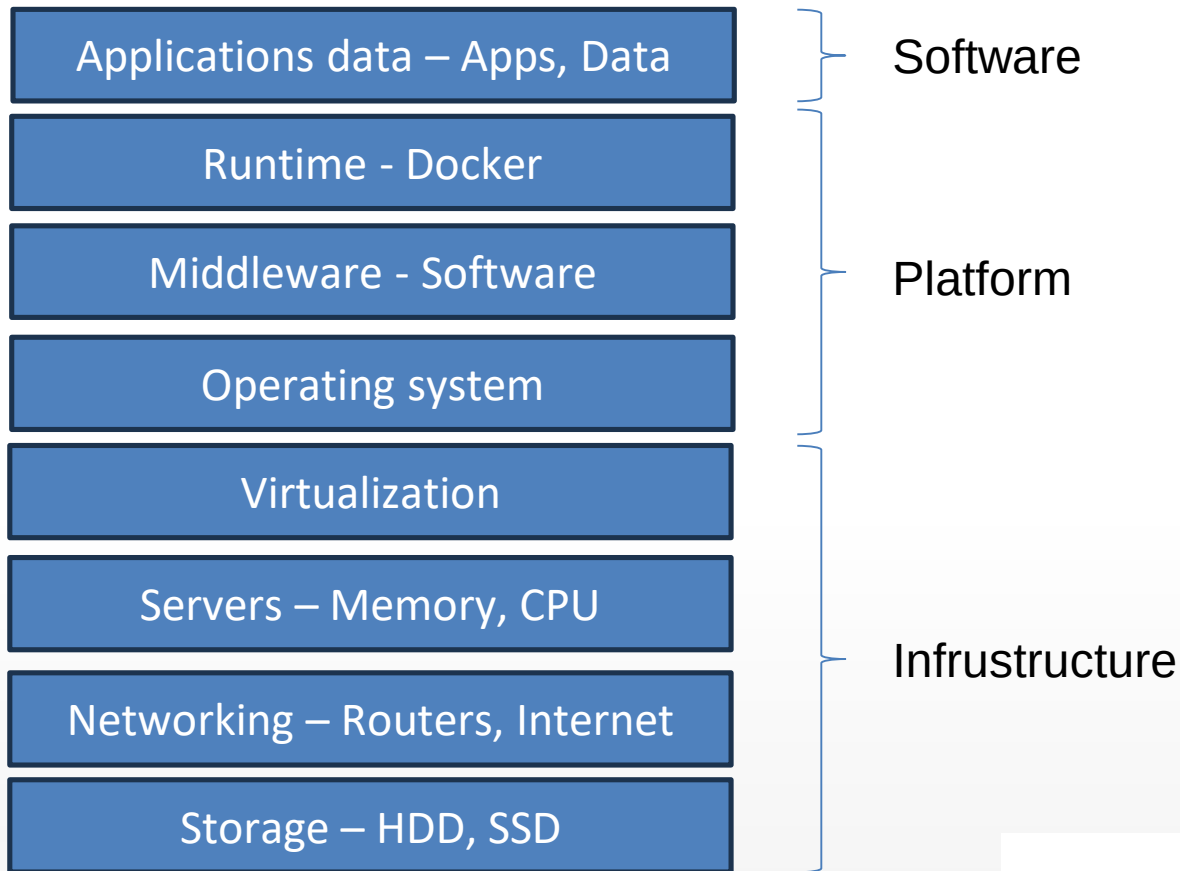
Operating system

Virtualization

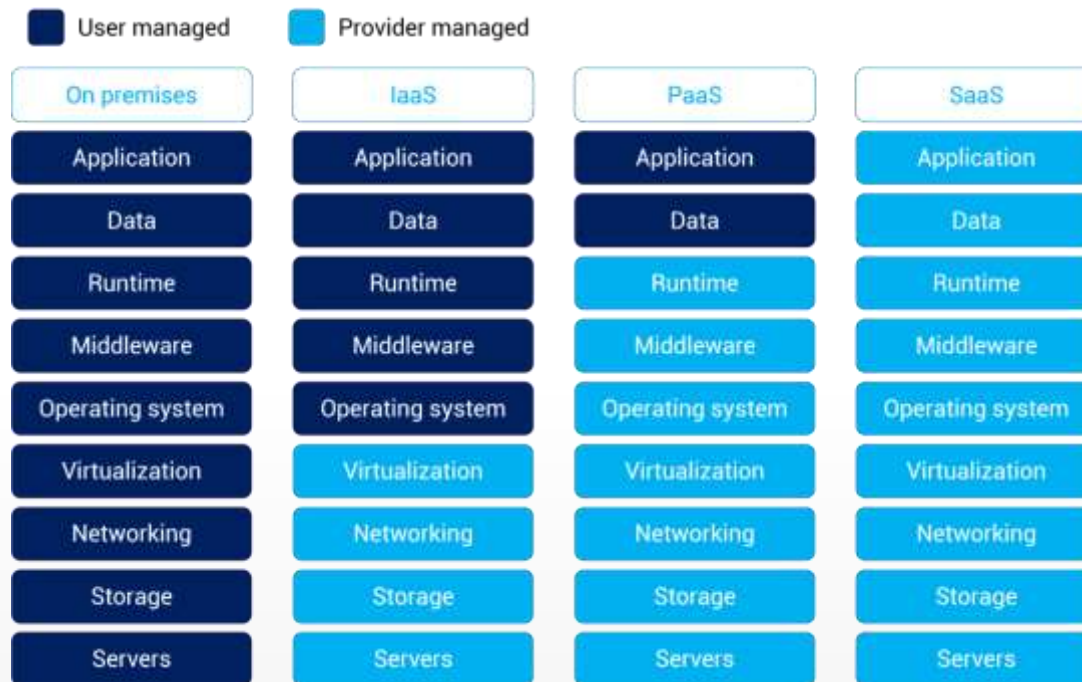
Servers – Memory, CPU

Networking – Routers, Internet

Storage – HDD, SSD



IaaS, PaaS and SaaS

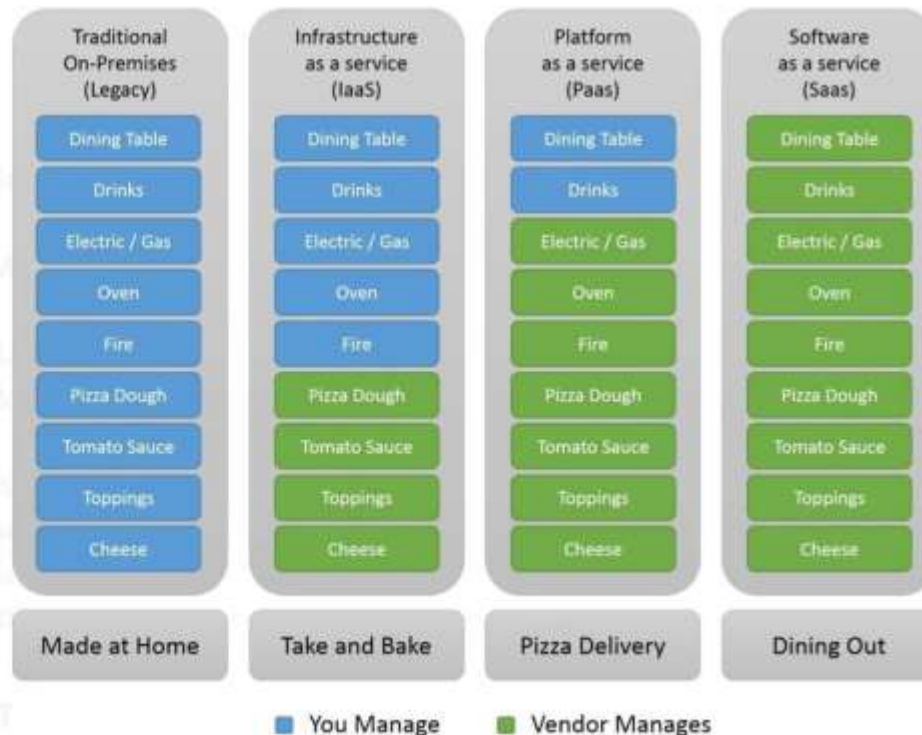


<https://www.alibabacloud.com/de/knowledge/what-is-paas>

Pizza as a service



<https://m.oursky.com/saas-paas-and-iaas-explained-in-one-graphic-d56c3e6f4606>



Benefits of IaaS

Business Advantages

- IaaS frees up human resources to let consumer focus on core businesses and content delivery rather than on IT infrastructure which does not generate business value.

Technical Advantages

- IaaS allows consumer to adjust processing capacity by scaling the infrastructure resources vertically and horizontally in an automated manner according to the change of processing demand.

Financial Advantages

- IaaS typically offers flexibility on financial model. The essence of IaaS financial model are Pay-As-You-Go and Subscription.

Pay-As-You-Go model is based on the consumption of resources such as per hour/day usage.

Subscription model is based on the period of resources reservation for the consumer and the price is fixed within the period regardless of the actual consumption.

<https://www.alibabacloud.com/de/knowledge/what-is-paas>

Benefits of PaaS

Business Advantages

- PaaS frees up human resources to let consumer focus on core businesses and content delivery rather than on platform software and its underlying IT infrastructure which does not generate business value.

Technical Advantages

- PaaS allows the consumer to adjust processing capacity by scaling up or down platform resources vertically and horizontally according to any changes in demand for processing power. In addition, provisioning a platform software could take a few weeks or even a few months. For example, installing an Apache Hadoop Filesystem and HBase could take several months.

Financial Advantages

- Same as IaaS, SaaS (*Pay-As-You-Go or Subscription*)

<https://www.alibabacloud.com/de/knowledge/what-is-paas>

Benefits of SaaS

Business Advantages

- SaaS shortens the application or software delivery time as it removes the need for architecture planning, software implementation, application customization, performance test, load test, etc.

Technical Advantages

- SaaS offers flexible scalability and high availability across different geographical locations. Traditionally, implementing enterprise software usually takes months from project start to production go live.

Financial Advantages

- Same as IaaS, PaaS (*Pay-As-You-Go or Subscription*)

But: Data security, privacy, data storage, sensitive data in other hands?

<https://www.alibabacloud.com/de/knowledge/what-is-paas>

Productized service

DEF

Productized services are standardized services sold like products, with clearly defined parameters and pricing.



From your customer's perspective, a productized service offers a specialized “done for you” solution with a compelling value proposition, packaged at a set price and scope.

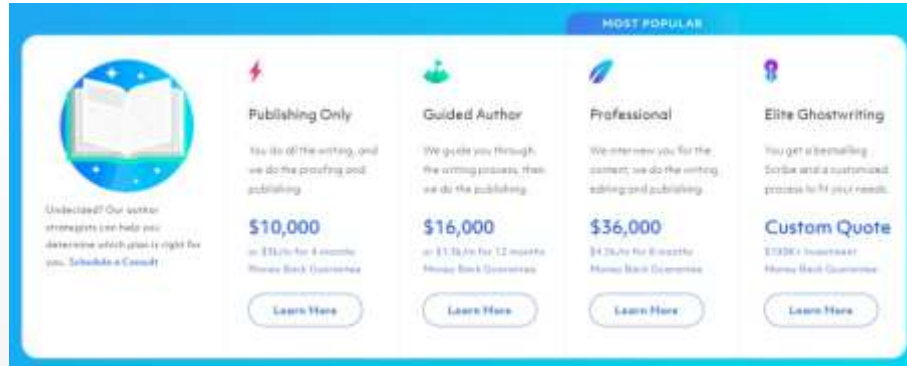
From the suppliers's perspective, a productized service is one that runs systematically and continues to produce and grow with or *without* your direct involvement.

S

Casel (2020)

Pitfalls of productized services?

Productized service: Example “Scribe”



There are two major benefits to productized services, and they are intertwined:

Scalability and leverage.

Having a properly productized service means it can be **scaled** far beyond you, and serve dozens, hundreds, thousands or tens of thousands of clients more than you could do yourself.

This is what creates the ultimate competitive advantage in life—**leverage**. You can do so much more with so much less when you can scale—assuming you do it properly.”

S Casel (2020)

Pitfalls of productized service

Interactivity / Uno-Actu principle

- The value that comes from the interactivity of a service is not or only to some extent existing anymore. The interaction during value creation is of high importance.

Individuality / less personalized service

- Productized services are less personalized than traditional service-based models, which can lead to less satisfied clients. Some clients will prefer to speak to service providers personally before making a purchase.

Narrower target market

- When you narrow your services down to something that can be neatly packaged and sold like a product, your pool of potential customers can become narrower too.

High initial investment

- As similar to a product, there is a need of high initial investment.

<https://manyrequests.com/blog/the-pros-and-cons-of-productizing-your-service/>

Chapter summary

- **Services** are immaterial and follow the uno-acto principle
- **SDL** states that all economies are service economies. Products mark services
- **Value** is the amount of progress a beneficiary feels having achieved
- **Value in use and value in interaction** as different forms of individual value
- **Modern technology enables software as a service and different provisioning models.**

